Kush Patel

Pittsburgh, PA

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Education

Carnegie Mellon University - GPA: 3.73/4.0 **Pittsburgh, PA**

Masters of Science in Artificial Intelligence & Information Security (Advanced Study) ***May 2027***

CourseWork: Machine Learning, Deep Learning, Advanced NLP, ML with Large Datasets, Trustworthy AI, Cloud Security, Distributed Systems, Secure Coding, Advanced Data Networks

Rutgers University - New Brunswick - GPA: 3.67/4.0

Bachelor of Science in Computer Science & Minor in Business Administration ***Sept 2022 – Aug 2024***

CourseWork: Design & Analysis of Computer Algorithms, Artificial Intelligence, Deep Learning, Operating Systems & Design, Systems Programming, Computer Security

Technical Skills

Languages: Python, Go, C/C++, React, TypeScript

Technologies: PyTorch, TensorFlow/Keras, Langchain, Spark, JAX, OpenCV, Polars, FastAPI, Flask, Yolo

Others: Postgres, BigQuery, VectorDB, Git, Docker, AWS, GCP, Ubuntu, VSCode

Experience

Advantest America, Inc **South Plainfield, NJ**

Software Engineer ***Jun 2024 – Present***

- Designed and optimized a low-latency computer vision and OCR pipeline, improving model accuracy by 13% and achieving sub-2 ms ONNX inference on live 4K video streams
- Built scalable ETL pipelines with Apache Spark and dbt, processing 500M+ records to power downstream ML and analytics workloads
- Integrated robotic APIs with legacy measurement software through a TCP/IP communication bridge, increasing automated inspection throughput by 90%
- Architected and deployed a scalable full-stack dashboard platform with FastAPI, caching, and optimized data processing for real-time applications

Technology Intern ***Aug 2022 – May 2024***

- Strengthened identity security for 500+ users through Azure AD IAM, MFA enforcement, and conditional access policies
- Automated audits and backup verification across 1000+ endpoints, streamlining compliance reporting and asset tracking

Projects

Multi-Layer Verification Prompt Injection Defense Pipeline | *Ollama, API, GCP* **Link**

- Designed a 3-layer inference-time defense pipeline (Structural Delimiters, Router LLM Firewall, and LLM-as-a-Judge) for LLM agents, cutting Direct Harm Indirect Prompt Injections on InjecAgent from 75.0% to 4.5% ASR.
- Implemented a deterministic symbolic verifier and TF-IDF RAG threat-intelligence module, intercepting 97.7% of residual LLM-judge failure cases and reducing projected attack success to 0.4%.

When to Stop Thinking: Adaptive Chain-of-Thought Reasoning in LLMs | *vLLM* **Link**

- Developed a lightweight MLP probe over 4,096-dimensional LLM hidden states to predict answer readiness at each reasoning step, enabling adaptive chain-of-thought early exiting with ∼0.80 ROC-AUC
- Built distributed inference and evaluation pipelines for long chain-of-thought workloads, achieving up to 98.1% simulated token savings across four reasoning benchmarks

GOD-100: Gap & Product Count Estimation in Dense Retail Environments | *DETR, Yolo, Pytorch* **Link**

- Engineered SKU-110K-S (algorithmic gap generation via UniK3D depth) and manually annotated GOD-100 (10k+ objects, 1.3k+ gaps with physical depth measurements) to benchmark retail shelf void estimation.
- Designed depth-fusion architectures (Faster-RCNN + UniK3D, FCOS + DA2) with custom regression heads to jointly detect shelf gaps and predict missing product counts, achieving 14.5 RMSE.

CMU & Pittsburgh QA RAG System | *Search Engines, Crawl, Selenium* **Link**

- Developed an end-to-end question-answering **RAG Model from scratch** on all-MiniLM-L6-v2 and re-rank on ms-marco-MiniLM-L-6-v2 for Pittsburgh and Carnegie Mellon University history, culture, trivia, and upcoming events

Lang IT | *Language Translation, GCP, VectorDB* **Link**

- Integrated real-time overlay translated text on live screen sharing along with **RAG** for IT suport from live screen